# Harsh Shroff

Open to Relocation

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## SUMMARY

**AWS Machine Learning Associate Certified** AI/ML Engineer and Data Scientist with hands-on experience delivering AI systems in sectors including geospatial intelligence, robotics and precision aquaculture. Specializing in generative AI, LLMOps, and agentic AI, with a portfolio of production-ready tools built using LangChain, Claude, YOLOv8, and Streamlit. Proven success in automating unstructured data pipelines, deploying real-time CV/NLP solutions, and coauthoring peer-reviewed research. Known for bridging the gap between experimentation and execution, translating complex models into scalable user-facing platforms that drive real-world impact. Ready to lead AI transformation efforts with a rare blend of deep technical fluency, strategic thinking, and an outcome-first mindset.

## SKILLS

**Programming:** Python, R, SQL, C++, Embedded C

**AI/ML & Data Science:** Generative AI (LLMs, RAG, LangChain, LlamaIndex), Deep Learning (PyTorch, TensorFlow), YOLOv8, OpenCV, Computer Vision (CV), NLP, Reinforcement Learning, Time Series, Predictive Modeling, MLOps (MLflow, Kubeflow), Prompt Engineering

**Robotics & Automation:** ROS, Robotics Integration (Boston Dynamics Spot, Clearpath Husky, Jackal), LiDAR, HoloLens 2, Sim2Real Deployment, Multi-modal Navigation, Active Perception

**Cloud/Data Engineering:** AWS (Lambda, EC2, S3, Textract, Comprehend, Glue, Redshift, Kinesis, Bedrock), Databases (PostgreSQL, MySQL, MongoDB, Elasticsearch), ETL, Airflow, Terraform, Spark, Hadoop

**Visualization & BI:** Tableau, Power BI, Streamlit, Dash, Seaborn, Matplotlib, Plotly

**Software Engineering:** API Dev (FastAPI, Flask), Docker, Kubernetes, Terraform, CI/CD, Git, Agile

## EXPERIENCE

### AI Researcher — Jun 2025 – Present

*Center for Real-Time Distributed Sensing and Autonomy (CARDS), UMBC*

- Developing a human-robot teaming system integrating Boston Dynamics Spot with real-time spatial mapping and augmented reality using Microsoft HoloLens, enabling Spot to precisely mimic human movements. Watch on YouTube

### ML Engineer — Jun 2024 – May 2025

*VITG Corp., Halethorpe, MD*

- Engineered a Llama LLM-based Slack bot with AWS Lambda, EC2, and S3 to automate candidate screening. This reduced process time by 60% and improved overall workflow efficiency.
- Built a geospatial chatbot tool using Claude 3.5 Haiku LLM and OCR tools like AWS Textract and LLMs, streamlining city zoning checks and decisions for planners in Hyderabad.
- Automated extraction from over 2,500 environmental documents with LLM-OCR pipelines. This improved data processing efficiency by 40% for regulatory and planning use.

### AI Researcher — Mar 2023 – Dec 2024

*Center for Real-Time Distributed Sensing and Autonomy (CARDS), UMBC*

- Led a team of 5 to build multi-modal robotic navigation for Boston Dynamics' Spot and Husky/Jackal robots (US Army/ArtIAMAS), pushing advances in sim-to-real navigation and autonomy for defense applications.
- Implemented gesture-based controls for robots with Microsoft HoloLens 2 and Python, enabling intuitive human-robot teaming for complex simulation environments, enabling the hands-free access for quadruped (legged) robots.
- Optimized LIDAR navigation with reinforcement learning, reducing navigation errors by 20%—improved real-time robotic pathfinding both in simulation and field tests.

### Data Science Intern — Aug 2023 – Dec 2023

*Freshwater Institute – The Conservation Fund, Shepherdstown, WV*

- Achieved 92% accuracy in fish fillet color grading using YOLOv8 and OpenCV models on Raspberry Pi, improving quality control for on-site aquaculture operations.
- Designed and prototyped an in-house, 3D-printed, portable grading kit for real-time fish fillet assessment.
- Co-authored a research paper on "Fish Fillet Color Profiling Using Deep Learning", highlighting deep learning's impact on aquaculture quality and food safety.

## EDUCATION

**Master's in Data Science** **Aug 2022 – May 2024**

*University of Maryland Baltimore County (UMBC)*

- GPA: 3.8/4.0
- Relevant Coursework: Natural Language Processing, Data Analysis & Visualization, Data Management, Big Data, Financial Data Science, Machine Learning, Artificial Intelligence

**Bachelor's in ELectronics & Communication Engineering** **Jun 2019 – May 2022**

*Gujarat Technological University (GTU)*

- GPA: 3.8/4.0 – *converted*
- Relevant Coursework: Introduction to Python, Introduction ML & AI, Internet of Things

## PROJECTS

**Real-Time Infrastructure Risk Assessment for Transportation Networks** (GitHub)

*Python, Flask, PostGIS, JavaScript, Leaflet, OpenStreetMap, OpenWeatherMap, TomTom Traffic API*

- Developed a full-stack web application to assess and visualize infrastructure risks for urban transportation networks in real time, integrating live traffic, weather, and spatial data on interactive Leaflet-based maps.
- Engineered scalable Flask APIs with PostgreSQL/PostGIS for fast spatial analytics, empowering city planners to filter, prioritize, and respond to roadway and network threats proactively based on dynamic multi-source risk scoring.

**Document Parsing for Environmental Planning** @ VITG Corp.

*AWS (Textract, S3, Redshift), Python, Tableau*

- Automated geospatial data extraction from 2,500+ scanned PDFs/images using AWS Textract, S3, Redshift, and Python, raising efficiency by 40%.
- Developed robust prompt-engineering pipelines and interactive Streamlit dashboards to support data-driven conservation project prioritization.

**Dashboard for Autonomous Robotic Patrols** @ CARDS, UMBC

*Python, ROS, LiDAR, Deep Learning, Computer Vision*

- Built a real-time asset monitoring dashboard with Python, ROS, LiDAR, and Dash, tracking battery health, connectivity, and GPS-based movement for robotic patrol simulations.
- Enhanced robotic safety and reliability by blending object detection via OpenCV with LIDAR-based obstacle avoidance in autonomous deployments.

**AirbnbLense – Personalized Recommendation System** (GitHub)

*Python, OpenAI API, Computer Vision, Tableau, Big Data*

- Engineered a cross-modal recommendation engine combining structured and unstructured data (OpenAI API, ML, Tableau), reducing bounce rate by 20% and raising engagement 40%.
- Implemented vision models for analyzing listing images on style and content, making recommendations that helped in improving overall user experience.

## ACHIEVEMENTS

### Certifications

- **AWS Machine Learning Associate Certification** – View Badge
- **AWS Machine Learning Foundations Certification** – Proficiency in core ML concepts & AWS tools.

### Publications

- **Ranjan, R.**, **Shroff, H.**, Sharrer, K., Tsukuda, S., & Good, C. (2024). *FilletCam AI: A handheld tool for precise fillet color profiling of Atlantic salmon and rainbow trout. Journal of Agriculture and Food Research, 18*, 101461. https://doi.org/10.1016/j.jafr.2024.101461
- **Trivedi, K.**, & **Shroff, H.** (2021). *Identification of deadliest mosquitoes using wing beats sound classification on tiny embedded system using machine learning and Edge Impulse platform.* In *2021 ITU Kaleidoscope: Connecting Physical and Virtual Worlds (ITU K)* (pp. 1–6). IEEE. https://doi.org/10.23919/ITUK53220.2021.9662116